

Theft Prevention

Group 21

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Introduction to Package Theft



Increasing Package Theft Rates

Package theft, or porch piracy, has surged by **36%** in recent years, affecting millions of homeowners and leading to significant financial losses.



Emotional Impact on Homeowners

Victims of package theft often report feelings of violation and insecurity, with **70%** stating it affects their sense of safety at home.



Limitations of Current Systems

Traditional security systems provide only passive monitoring, requiring homeowners to review footage after incidents, which is often too late to prevent theft.

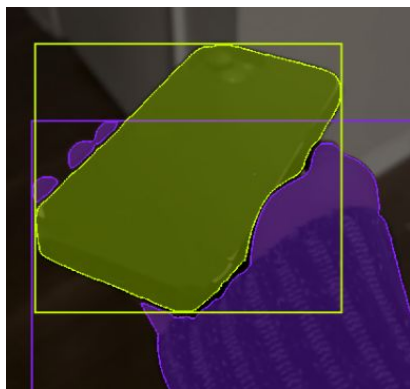
Our Solution:

Real-Time Package Theft Alert System

This presentation introduces an innovative **Real-Time Package Theft Alert System** that leverages advanced computer vision and deep learning technologies. By detecting suspicious activities in real-time, this system not only alerts homeowners but also captures critical evidence, ensuring a proactive approach to security.



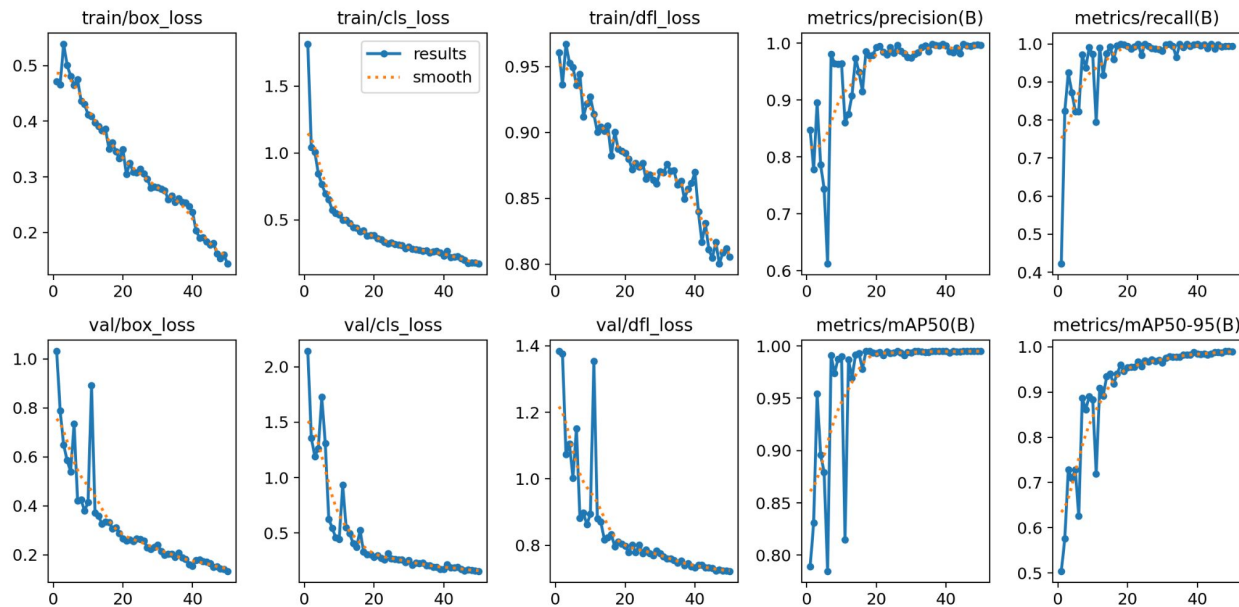
Data



- Collected real-world **video** data
- Extracted **training frames** from videos by converting video sequences into image datasets using **Ezgif** and **Roboflow** preprocessing tools
- **Annotated** objects using **Roboflow** labeling platform, ensuring consistent class definitions
- Structured dataset for model training by programmatically splitting into **train/validation** sets using a custom **Python** pipeline



Model Training



- Training losses **decrease** steadily, showing stable convergence
- Validation metrics (precision, recall, mAP) **reach near 1.0**, indicating strong performance
- Validation performance mirrors training, showing **effective learning**

Monitoring Conditions



Object Detection

The system identifies when the designated object is placed in it's view, marking it as 'safe' or 'at risk' for further monitoring.



Interaction Monitoring

Proximity of individuals to the object is closely monitored. If a person enters the object's area, the system flags this interaction for potential theft analysis.

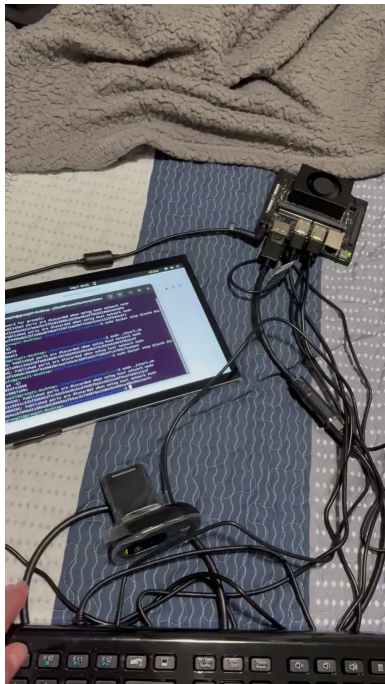


Theft Detection Logic

The system combines object placement and person proximity data. If a object disappears shortly after a person interacts with it, a potential theft is logged for review.



Jetson Demo Video





Future Work



Frontend Upgrades:

- Mobile device compatibility with customizable UI
- Push notifications

Backend Upgrades:

- Expansion to more objects (bikes, food deliveries etc.)

